# Engineering roofing granules with reactive mineral coatings to capture atmospheric $CO_2$

## Introduction

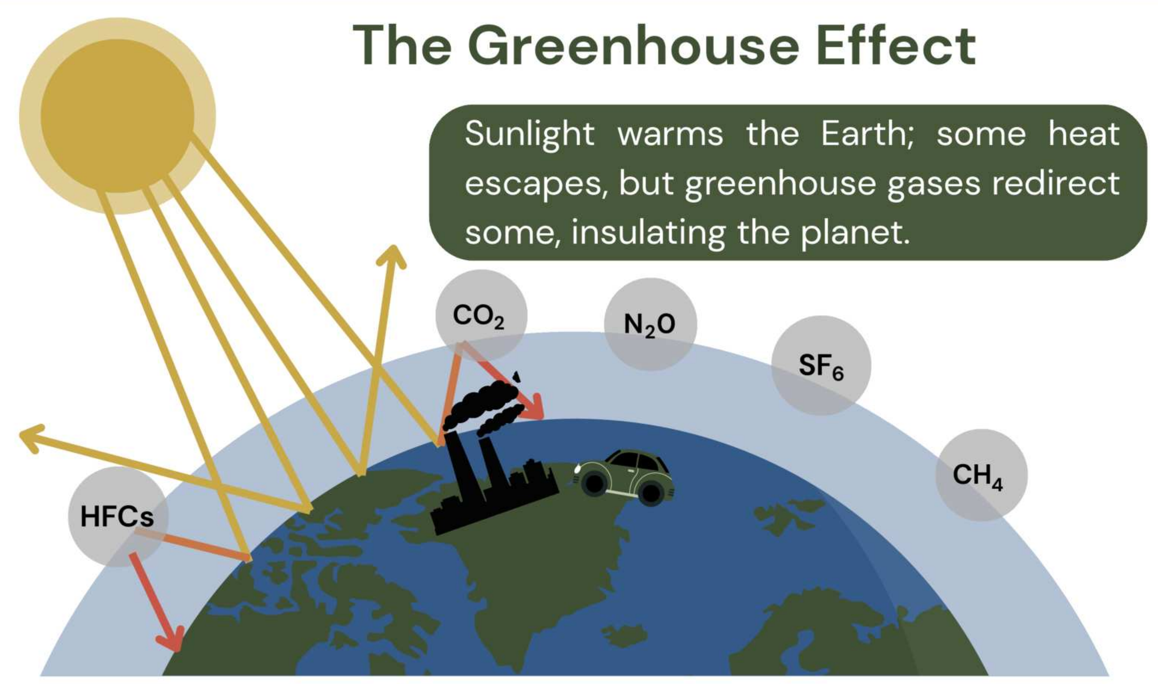


Figure 1: $CO_2$ emissions from factories

- Carbon dioxide ($CO_2$) levels have risen: Marking the highest concentrations in at least two million years (Erans et al., 2022)
- Two reduction forms mitigate carbon dioxide levels:
  - Eliminate from the source
  - Capture existing carbon dioxide emissions

- Emission reduction by itself is not enough to meet international climate goals
- This gap has led to growing interest in negative-emission technologies, especially direct air capture (Erans et al., 2022; Ozkan et al., 2022)
- Scalable and efficient carbon capture solutions are needed to address the impacts of climate change

Figure 2: Direct air capture and storage plant

## Purpose

- Carbon capture roofing granules are explored to reduce atmospheric $CO_2$: a novel approach
- Roofing materials coated with reactive minerals may capture carbon dioxide
- Roofing granules are durable, widespread, and already integrated into infrastructure
- If engineered to capture carbon, roofs could function as passive carbon sinks

### Groups

Control

| | |
|---|---|
| $Ca(OH)_2$ over 48 hours | $Mg(OH)_2$ over 48 hours |
| $Ca(OH)_2$ over 168 hours | $Mg(OH)_2$ over 168 hours |
| $Ca(OH)_2$ over 336 hours | $Mg(OH)_2$ over 336 hours |

## Methods

**01 Prepare Samples with mineral coatings**

- Prepare $Ca(OH)_2$ and $Mg(OH)_2$ mineral slurries

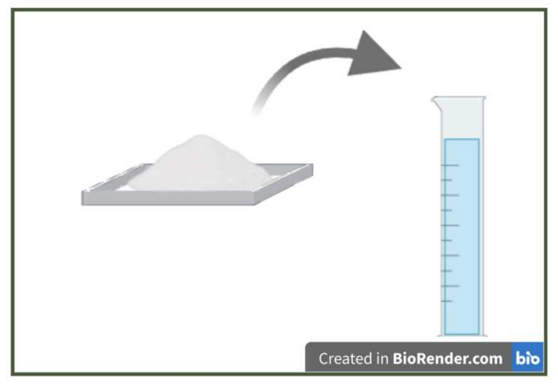


Figure 1: Calcium hydroxide being added to graduated cylinder of deionized water

- Soak roofing shingles in slurries


Figure 2: Hot air oven

- Dry in oven at 60°C for 48 hrs

**02 Pump $CO_2$ in BioChamber and disconnect**

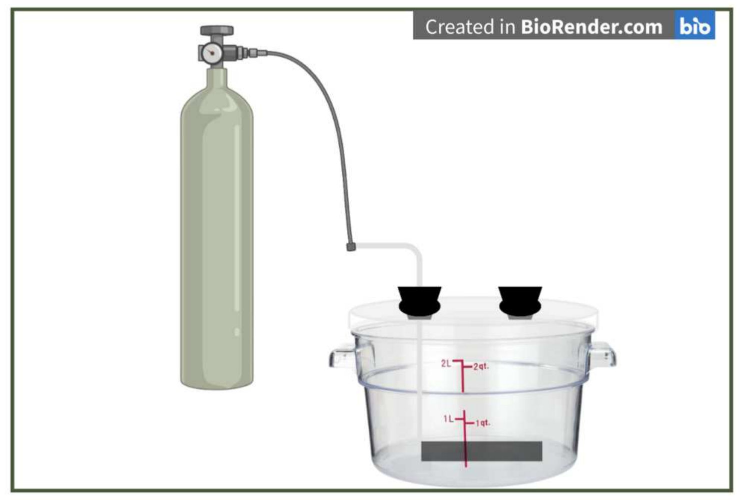


Figure 3: BioChamber attached to $CO_2$ gas chamber

**03 Gravimetric Carbonation**

- Record initial mass of samples
- Let trial run for durations of 48 hrs, 168 hrs, and 336 hrs
- Record final mass of samples

**04 Post-exposure processing**

- Collect condensation/rinse water
- Test alkalinity for carbonate presence
- Record surface and physical changes

**05 Repeat steps 1–5 for 3 replicates per treatment group**

- $Ca(OH)_2$-coated
- $Mg(OH)_2$-coated
- Uncoated control

Figure 4: Roofing Shingle sample coated in $Ca(OH)_2$

**06 Compare controls and groups using one way ANOVA**

## Results

### Box and Whiskers Plot

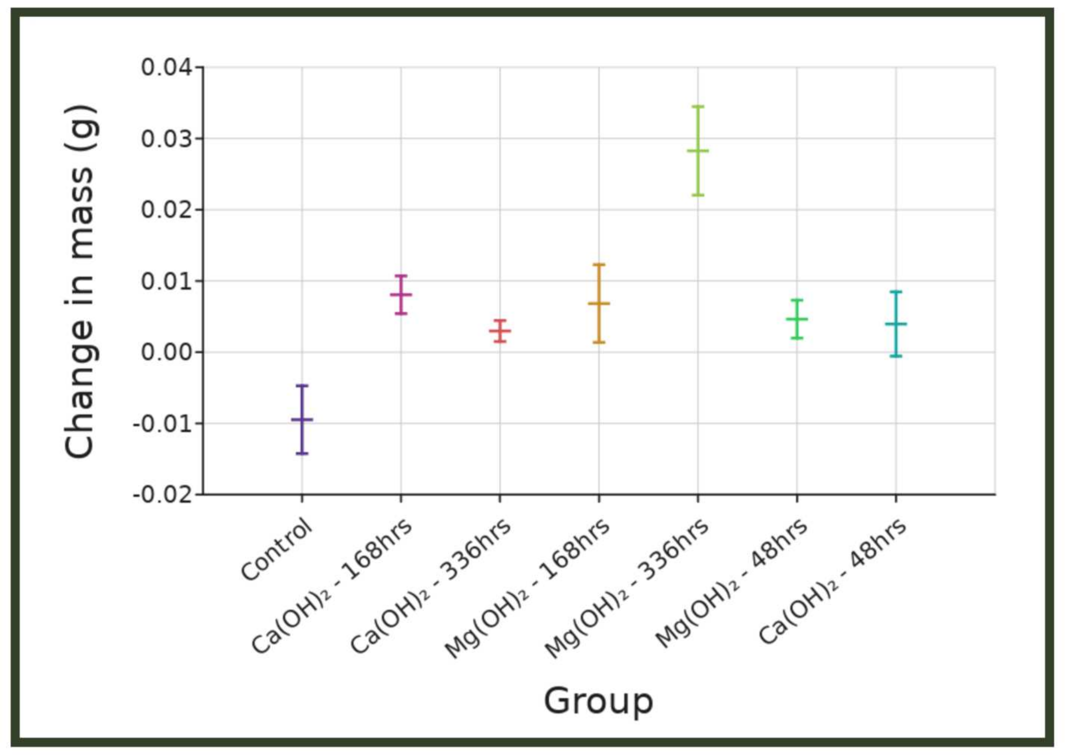


Figure 8: This graph displays the distribution of change in mass for each experimental group, including the control, $Ca(OH)_2$ at 48, 168, and 336 hours, and $Mg(OH)_2$ at 48, 168, and 336 hours.

### ANOVA Test

| Effect | Degrees of Freedom (df) | F-statistic (MS / MS residual) | P-value | Interpretation of P |
|---|---|---|---|---|
| Group | 4 | 37 | <0.01 | A P-value of **<0.01** means **extremely strong confidence** that the groups are different. |
| Error or Residual | 10 | | | |

Figure 9: This one-way ANOVA analysis evaluates whether differences in mass change among the experimental groups are statistically significant. A result of $p < 0.05$ indicates that at least one group mean differs significantly from the others.

| Pair | P-value |
|---|---|
| $Ca(OH)_2$ - 168hrs - Control | <0.01 |
| $Ca(OH)_2$ - 336hrs - Control | 0.01 |
| $Mg(OH)_2$ - 168hrs - Control | <0.01 |
| $Mg(OH)_2$ - 336hrs - Control | <0.01 |
| $Mg(OH)_2$ - 48hrs - Control | 0.01 |
| $Ca(OH)_2$ - 48hrs - Control | 0.01 |

### Post-hoc Tukey Test

Figure 10: A post-hoc tukey test was run to determine which specific groups differed from each other. Groups either varied very significantly ($p = 0.01$) or extremely significantly ($p < 0.01$), showing how observed differences were the result of the treatment.

## Limitations

- Time constraints limited the number of trials conducted
- Environmental conditions were not fully isolated
- Variations in coating thickness, sample preparation, and material distribution
- The experiment was performed under controlled laboratory conditions
- Actual roofs experience varying atmospheric $CO_2$ concentrations and weathering over long periods
- The model demonstrates the potential for passive carbon capture, but further large-scale and long-term testing is needed to determine practical effectiveness, durability, and economic feasibility in real-world applications

## Conclusion


- $Ca(OH)_2$ and $Mg(OH)_2$-coated roofing granules showed measurable increases in mass, suggesting successful carbonation and atmospheric $CO_2$ capture.
- Longer exposure times
- generally resulted in greater mass changes, indicating that carbonation continued over time.
- Statistical analysis (one-way ANOVA and Tukey test) showed significant differences between treatment groups and the control, supporting the effectiveness of reactive mineral coatings.
- These findings suggest that roofing materials coated with reactive minerals could serve as a scalable passive carbon capture technology.

## References


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